

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 16: General Equilibrium, Time, and Uncertainty

Notes on the Chapter

This chapter concerns the adaptation of General Equilibrium to economies with time and uncertainty. The plot, roughly, is

- Time and uncertainty are modeled with the usual event-tree structure, here called a time/uncertainty/information (t/u/i) structure. Only finite and discrete time and finitely many states of nature are considered. All information is held in common.
- In the classic manner of Debreu, an all-markets-at-once economy is set up in which commodities are contingency-stamped, reflecting the time and state of information in which the commodity is consumed/used as input/produced.
- Then, following Arrow, we suppose that instead of all markets at once, there are spot markets in each contingency and a full set of markets (at the outset) for transferring wealth back and forth from each contingency to the initial contingency. This is done in detail for contingent claims that pay off in some physical good, and then discussed (with details left to students, but provided in the *Guide*) for financial claims. The “equilibrium-outcome” equivalence of this economy with all-markets-at-once is stated and proved. In this section firms are suppressed; only pure-exchange is considered. (Firms continue to be suppressed until the final section.)
- More complex securities are considered. This is done at first for financial securities only, that is, those that pay dividends in the numeraire. Connections between no-arbitrage, model viability, and the existence of strictly positive “pricing vectors” (vectors normal to the subspace of feasible wealth transfers) are made. The complete-markets theory in this setting is thoroughly analyzed, with emphasis on the (unique for complete markets) pricing vector and its use in pricing complex securities as the “properly discounted present worth of future dividends.” The equivalence (for some sets of securities) of equilibria of plans, prices, and price expectations (EPPPE) and Walrasian equilibria is developed. The massive indeterminacy in

securities prices (because of the ability to rescale spot-market prices) is discussed, and the theory is recast in the context of securities that pay their dividends in a single commodity whose spot-market price is always chosen to be 1, which resolves the indeterminacy.

- Incomplete markets are (barely) introduced by way of Hart's classic example of two Pareto-ordered equilibria.
- And then firms are brought back in: If markets are complete regardless of what firms do and if they are price-takers in the full sense of the word, everything works like a charm. An example is then given to show that things go awry if these assumptions are not met: Shareholders want, *ex ante*, a firm to destroy valuable goods, although once trading takes place on that basis, the new shareholders want the firm to cancel those plans.

I've said elsewhere in this *Manual* that the book is one to two chapters too long for Stanford's ten-week Autumn quarter. In particular, until 2011, I *never* had enough time even to begin to cover this material. (In 2011, for the first time, I made time by compressing other material and by omitting comparative statics.) And, for a welcome change for me, but perhaps not for the reader, this is a topic that I know something about. Moreover, when I began to write this chapter, I wondered how I'd manage to stretch it to 15 pages. Just writing down the "obvious" stuff, and (due in part to length) suppressing all systematic treatment of incomplete markets, whether with or without firms, I am well over 50 pages. My lack of experience in teaching this material, combined with my personal involvement in its development (which, presumably, anchors my perceptions about what is easy and what is important) and with my lack of initial calibration on how long it would be, all these contribute to a nagging worry about the quality of exposition in this chapter.

I do have another excuse for suppressing systematic development of incomplete markets, besides length (and exhaustion): I more or less say this right at the end, but if markets are incomplete, the obvious question is, Why? And answers to that question seem to me to require methods and models that this volume has systematically eschewed. So, thematically, incomplete markets didn't seem to me to belong; it will be a worthwhile topic for Volume II, if Volume II ever makes an appearance.

As a final note on the construction of the chapter, it is worth pointing out that Section 16.3, on economies with simple Arrow-Debreu contingent claims, could be subsumed in the more general Section 16.4, with general securities markets (but no firms). I have kept a separate section on A-D contingent claims on two grounds: Some instructors may prefer to cover the basics only. And the notation and accounting (not conceptual) complexity of general securities markets may overwhelm first-time readers; the simple A-D contingent claim structure is probably helpful for building intuition before plunging into the notation hell of Section 16.4.

As already noted, my experiences in teaching this material is limited to one time (as I write this). On the other hand, and long ago, I had the experience of giving semi-

nars on some of these topics. Based on that limited experience, it seems to me that the real problem for students with this material will be that the “indirect” way of constructing dynamic equilibria (using π) is hard to appreciate from just reading propositions about it. So, when I did teach this, I spent a few minutes at the end of one class introducing the basics of a time/uncertainty/information structure and then adapting Walrasian equilibria to contingency-stamped commodities (that is, I covered Sections 16.1 and 16.2). Then, on the day we discussed the Chapter, I put the focus on Problem 16.9 as a motivating/illustrative example. (Students were told to prepare this problem carefully.) I’m not certain, but I think it worked. Note that this problem is not solved in the *Student’s Guide*, so students cannot easily “read” the solution and think they know what is going on. (The early parts of the problem—finding the unique Walrasian equilibrium and proving it is unique—was harder than I anticipated, but it did provide a nice review of all sorts of things covered during the term. But because this took longer than anticipated, I never got to any discussion of firms or the many pathologies of incomplete markets.)

Notes on the Problems

Again noting my lack of experience teaching this, I reiterate that Problem 16.9 provides a relatively concrete example of a lot of what happens in this chapter and so is a good problem to assign.

Beyond that problem, it seems possible to me that students will get lost in the “bookkeeping” that goes on, as one moves back and forth between all-markets-at-once and the more dynamic-market economies that are the bulk of the chapter. The bookkeeping isn’t hard, but until you sit down and do it once or twice, it seems so. So I’ve tried to write in a way and give problems that force students to confront the bookkeeping issues. In particular, I think Problem 16.2 is a useful, albeit tedious, starting exercise.

In my opinion, the key results in Section 3 are Proposition 16.8 and then Lemma 16.11. Going step by step through Problem 16.5 and, in particular, proving the first part by an arbitrage argument and then with π is a good idea. That said, it is a good idea that I personally have not had the time to try.

Solutions to Non-Starred Problems

■ 16.1. (a) Propositions 3.1 tells us that budget constraints all hold with equality; Proposition 14.4(d) tells us that equilibrium prices are nonnegative. Since equilibrium prices are nonnegative and all consumers are locally insatiable, the First Theorem of Welfare Economics, Proposition 15.2, gives Pareto efficiency.

Now fix a contingency f_t and let h be the consumer whose preferences for f_t -contingent wheat (commodity 1) are globally insatiable. If we knew that the distinguished consumer has strictly increasing preferences in f_t -contingent wheat, Proposition 14.5 could be invoked to tell us that the price of f_t -labelled wheat is strictly positive. But we

don't know this, so we need to work a bit (but only a bit). Suppose that $p_{1f_t} = 0$ in a Walrasian equilibrium. (We know that it can't be strictly negative by Proposition 14.4(d).) Then our distinguished consumer could, from any bundle, move to a bundle with as much contingency- f_t wheat as she desires; it is free. The assumption says that with enough extra wheat, her utility is strictly increased. That would be incompatible with being in a Walrasian equilibrium.

(b) Suppose (r, q, x, y) is an EPPPE and, for some basic commodity i and contingency f_t , $r_{if_t} < 0$. By Assumption 16.1', some distinguished consumer is globally insatiable in contingency- f_t consumption; call this consumer h . So whatever is x^h , there is some $\hat{x}$ such that $\hat{x}$ is identical to x^h except on f_t , and $u^h(\hat{x}) > u^h(x^h)$. Of course, h may not be able to afford $\hat{x}_{f_t}$ in contingency f_t . But by increasing her consumption of i in contingency f_t , h can loosen her contingency- f_t budget constraint by any amount she wishes, since the price of contingency- f_t labelled i is negative. And since her preferences are nondecreasing, doing this does not diminish her utility. So if $\hat{x}$ is not affordable for her in contingency f_t (since it is the same as x^h in all other contingencies, it is affordable in all other contingencies), some $\check{x}$ which is the same as $\hat{x}$ but with more contingency- f_t commodity i is affordable and, of course, satisfies $u^h(\check{x}) > u^h(x^h)$. This contradicts the "utility maximization" part of the definition of an EPPPE, a contradiction.

So $r \geq 0$.

(c) Suppose that $r_{1f_t} \leq 0$ in an EPPPE. The argument just given in part (a) says that, from any alleged equilibrium consumption allocation x^h , our distinguished consumer can increase her consumption of wheat without bound while staying within her budget constraint, which (eventually) means an increase in her utility. This is incompatible with being part of an EPPPE.

Now consider the specific dynamic economy of Section 16.3 with a full set of contingent futures markets, and suppose that in an EPPPE, $q_{f_t} \leq 0$ for some f_t . The distinguished consumer can then increase her consumption of f_t wheat without bound by buying futures in f_t wheat; if $q_{f_t} = 0$, this has no effect on her f_0 budget constraint, while if $q_{f_t} < 0$, it loosens that constraint. This is also incompatible with being part of an EPPPE.

■ 16.3. If U is globally insatiable in each x_t and $\mu(\omega) > 0$ for all $\omega \in \Omega$, then Alice is globally insatiable in each f_t : Choose any f_t and any bundle $x \in R_+^{kN}$. Let ω be a state contained within f_t , and let $(x_0(\omega), \dots, x_T(\omega))$ be Alice's consumption path in state ω as per x . Of course, $x_t(\omega) = x_{f_t}$. Let $x'_t \in R_+^k$ be such that

$$U(x_0(\omega), \dots, x_T(\omega)) < U(x_0(\omega), \dots, x_{t-1}(\omega), x'_t, x_{t+1}(\omega), \dots, x_T(\omega)).$$

Create $x' \in R_+^{kN}$ by $x'(f'_t) = x(f'_t)$ for all $f'_t \neq f_t$, and $x'(f_t) = x'_t$. Then the expected

utility of x' less the expected utility of x is precisely

$$\sum_{\omega \in f_t} \mu(\omega) \times \left[U(x_0(\omega), \dots, x_{t-1}(\omega), x'_t, x_{t+1}(\omega), \dots, x_T(\omega)) - U(x_0(\omega), \dots, x_T(\omega)) \right],$$

which is strictly positive by design.

Note that this characterization of f_t global insatiability is, in fact, if and only if: Suppose for some $\hat{x} = (\hat{x}_0, \dots, \hat{x}_T) \in R_+^{k(T+1)}$, we have $U(\hat{x}) \geq U(\hat{x}_0, \dots, \hat{x}_{t-1}, x_t, \hat{x}_{t+1}, \dots, \hat{x}_T)$ for all $x_t \in R_+^k$ (that is, starting from $\hat{x}$, no change in the time- t consumption will increase utility). Then if $x \in R_+^{kN}$ is $x(f'_{t'}) = \hat{x}_{t'}$ for all t' and $f'_{t'} \in \mathcal{F}_{t'}$, no change in x_{f_t} (alone) can increase Alice's expected utility. And if $\mu(\omega) = 0$ for some ω , then Alice's expected utility of x is unaffected by changes in $x_{\{\omega\}}$.

■ 16.4. It is not true that, in a general EPPPE, all the budget constraints of each consumer hold with equality, even if Assumption 16.1' holds. To see this, consider a one-basic-good, one-consumer economy, with one state of nature and two dates, $t = 0, 1$. There are no securities, so the "equilibrium" amounts to separate spot-market equilibria in each of the two spot markets. The one consumer's preferences are given by

$$u(x_0, x_1) = \min\{x_0, 5\} + x_1.$$

Her endowment is 10 units of the good at date 0 and 1 unit at date 1. This consumer's preferences are clearly nondecreasing and locally insatiable; wherever she is, she gets more utility by increasing x_1 .

But one EPPPE for this economy is given by spot market prices of 1 at each date, and the two-period consumption plan (5, 1) for the consumer. In the date 0 spot market, her wealth is 10, but she is content (satiated) to spend 5, while she is behaving entirely "normally" at date 1. (It is true that another EPPPE has her consume (10, 1), and in that equilibrium her individual budget constraints all hold with equality; to get an example where the budget constraints are never equations in any EPPPE, we would need consumers whose preferences are not nondecreasing, and we'd need at least two consumers. Can you fill in the details?)

What does it take to get consumers to satisfy each budget constraint with equality in every EPPPE? There are two paths to this.

- If we know that markets are complete, we will know (given Assumption 16.1') that every consumer satisfies every budget constraint with equality: If some budget constraint were slack, the consumer could use the securities markets to transfer her "excess" wealth in that state to some other contingency where she can use the extra wealth; we know that there must be such a contingency, because she is (overall) locally insatiable.

- Or this follows (by the standard argument) if the consumer is locally insatiable in each contingency, meaning that from any consumption plan x and for every $\epsilon > 0$ and contingency f_t , there is better (utility increasing) plan x' that (a) lies within ϵ of x and (b) agrees with x on all contingencies except for f_t .

■ 16.7. Suppose S and q are viable. Fix any contingency f_t for $t < T$, and enumerate its $\ell(f_t)$ immediate successors as $f_{t+1}^1, \dots, f_{t+1}^{\ell(f_t)}$. We are told that there are $\ell(f_t)$ securities that trade at f_t , pay dividends (only) at the immediate successors to f_t , and whose dividend patterns are linearly independent. Enumerate these as $s^1, \dots, s^{\ell(f_t)}$. (Note that this enumeration is done only after f_t has been fixed. For a different contingency, an altogether different security would be superscripted with 1, and so forth.) Since the patterns of dividends they pay are linearly independent, they span $R^{\ell(f_t)}$; in particular, for each f_{t+1}^j a portfolio $\alpha^j \in R^{\ell(f_t)}$ of these $\ell(f_t)$ securities can be formed so that

$$\sum_{i=1}^{\ell(f_t)} \alpha_i^j d_{s^i}(f_{t+1}^n) = \begin{cases} 1, & \text{if } n = j, \text{ and} \\ 0, & \text{if } n \neq j. \end{cases}$$

In other words, the j th portfolio delivers 1 unit of whatever the dividends are paid in (numeraire for Section 16.4, and wheat for Section 16.5) if the next contingency is f_{t+1}^j and 0 if it is any other immediate successor of f_t . I know that this portfolio has a positive net cost to purchase, because it would otherwise represent an arbitrage opportunity. This shows how to move wealth from any f_t to any of its immediate successors; by stringing these together in the fashion of the solution to Problem 16.6 (given in the *Guide*), we can move wealth from f_0 to any f_t , and Lemma 16.10 applies.

■ 16.9. (a) In the all-markets-at-once economy, any equilibrium must be Pareto efficient. Since the social endowment is constant at each date—2 units at date 0, 8 units at date 1, and 18 units at date 2—and since both Alice and Bob are strictly risk averse, efficiency requires that the consumption levels of each consumer at each date is constant: If it were not, giving each consumer her or his average consumption in all contingencies associated with a given date would be a Pareto improvement (and is clearly feasible). So the only consumption allocations that are candidates for Walrasian-equilibrium outcomes give to Alice for her consumption some x_0^A in f_0 , x_1^A in both time-1 contingencies, and x_2^A in the four time-2 contingencies. But this then requires that the prices of the good in the two time-1 contingencies must be the same; as otherwise Alice would not be equating her bangs-for-the-buck across those two contingencies. And if the prices in the two time-1 contingencies are the same, then the value of Alice's endowment is the same as Bob's, so (because the utility functions are identical and strictly concave) Alice and Bob must be consuming the same amounts.

Hence the only Walrasian-equilibrium consumption allocation is where Alice and Bob both consume 1 unit in f_0 , 4 units in each of the time-1 contingencies, and 9 units in each of the four time-2 contingencies. Prices must be proportional to their marginal

utilities: If we normalize p_{f_0} to be 1, giving a bang-for-the-buck in f_0 of 1, then $p_{f_1^1} = p_{f_1^2}$ must be such that

$$\frac{1}{2} \frac{1}{4p_{f_1^1}} = 1,$$

or $p_{f_1^1} = 1/8$, and $p_{f_1^2} = 1/36$ for each of the four time-2 contingencies. To be pedantic about this, if Alice (say) consumes x_1 in each of the time-1 contingencies, then the portion of her utility function that pertains to contingency f_1^1 is $(1/2)\ln(x_1)$, where the $1/2$ is the probability of this contingency. (Remember, both Alice and Bob are expected utility maximizers.) Hence her marginal utility is $1/(2x_1)$. In equilibrium, $x_1 = 4$, so her marginal utility for $x_{f_1^1}$ in equilibrium is $1/8$, and the price of consumption in that contingency must make the bang-for-the-buck 1, which is $p_{f_1^1} = 1/8$.

Therefore, listing prices and then consumption allocations in the order f_0, f_1^1, f_1^2 , and then the four contingencies for time 2, the unique Walrasian equilibrium with all markets operating at once is $p^* = (1, 1/8, 1/8, 1/36, 1/36, 1/36, 1/36)$ (or something proportional to this), $x^{\text{Alice}} = (1, 4, 4, 9, 9, 9, 9)$, and $x^{\text{Bob}} = (1, 4, 4, 9, 9, 9, 9)$.

(b) Now we are looking for EPPPE for the two securities provided and for $\pi = (1, 1, \dots, 1)$. Having specified a π , we can compute q by formula (16.4), getting

$$q_{f_0} = (17, 22), q_{f_1^1} = (9, 14), \text{ and } q_{f_1^2} = (8, 8),$$

where I'm listing for each contingency the price of the first asset and then the price of the second asset. Note that starting from f_0 , the vector of "immediate values" of the two securities are linearly independent, which means that a portfolio of them can be constructed at time 0 that brings wealth 1 to contingency f_1^1 and wealth 0 to f_1^2 , while another portfolio will bring wealth 1 to f_1^2 and 0 to f_1^1 . (Moreover, I *know* that these two portfolios will each cost 1 at time 0. If you don't see why I know this, think about it more.)

Of course, the dividend vectors in the two successors to f_1^1 , namely $(2, 12)$ and $(7, 2)$, are linearly independent, and the dividend vectors in the two successors to f_1^2 , namely $(3, 4)$ and $(5, 4)$, are linearly independent. So with this choice of π , $M(S, q)$ will have dimension 6; markets will be complete. Therefore, we know that this q , together with $r \equiv p^*$, the Walrasian-equilibrium outcome, and appropriate trading plans (which we *know* must exist, even if we don't bother to compute them), constitute an EPPPE. (Exactly which Propositions tell us that this is true?)

Although we don't need to find the equilibrium trading plans that complete the EPPPE, it might be good practice to do so; here are the computations for Alice: She must move 2 units of consumption from f_1^1 (where her endowment is 6) to f_1^2 (where it is only 2) and, since spot prices in the two time-1 contingencies are $1/8$ apiece, this means she must move $1/4$ unit of wealth from f_1^1 to f_1^2 . She needs to find a portfolio of the two

assets that does this: Letting α_1 be her initial holdings of the first asset and α_2 her holdings of the second asset, her trading strategy (buy the portfolio α at time 0 and sell it at time 1) must satisfy

$$9\alpha_1 + 14\alpha_2 = -1/4, 8\alpha_1 + 8\alpha_2 = 1/4, \text{ and } 17\alpha_1 + 22\alpha_2 = 0.$$

(Bob, of course, will take the opposite position.) That is three equations in two unknowns, but one of the equations is redundant (add the first two and see what you get), and the solution is $\alpha_1 = 11/80$ and $\alpha_2 = -17/160$.

(c) Now suppose that instead of being financial, the two securities paid their dividends in the one consumption good. To nail things down (really, to ensure that we don't need to multiply dividends by a spot price in the budget constraints), we normalize prices in each spot market so that the spot price of the (one!) good is always one. Now, if we are going to replicate the Walrasian equilibrium, π has to pick up the relative values of the consumption good in different contingencies; we need $\pi_{f_t} = p_{f_t}^*$, where p^* is the equilibrium price vector in the Walrasian equilibrium. Applying formula (16.4) to find q , we get

$$q_{f_0} = (17/36, 22/36), q_{f_1^1} = (2, 28/9), \text{ and } q_{f_1^2} = (16/9, 16/9).$$

(If you aren't clear on these numbers, get clear!) The "immediate returns" vectors from time 0 to time 1 are linearly independent, so we know we'll have complete markets. So the answer to the question posed is, Yes, the Walrasian-equilibrium outcomes are EPPPE outcomes, going with $r \equiv 1$, q as given above, and some y that I'll leave for you to compute if you must.

(d) Now we return to the case of financial securities and, specifically, what happens if we take

$$\pi_{f_t} = \begin{cases} 1/2, & \text{if } t = 2 \text{ and } f_t = \{\omega_1\}, \text{ and} \\ 1, & \text{for all other contingencies.} \end{cases}$$

Using (16.4) once again, this gives prices for the two securities of

$$q_{f_0} = (16, 16), q_{f_1^1} = (8, 8), \text{ and } q_{f_1^2} = (8, 8).$$

The time-0 to time-1 returns on the securities are co-linear, and there is no way to transfer wealth between the upper and lower halves of the t/u/i tree, which is precisely what needs to be done to get to the efficient, Walrasian-equilibrium outcome.

Is there an EPPPE for this π ? There is, but we'll need to sweat a bit for it. Alice and Bob can use the securities to effect some "smoothing" of their consumption between times 1 and 2 in the two halves of the t/u/i tree. In the top half, Bob is richer, so he'll

be happy to give Alice some time 1 consumption in return for some time 2 consumption, in a ratio that will also be agreeable to Alice. In the bottom half, the reverse is true. Since "markets are complete" in the top half and in the bottom half, you can look for Walrasian equilibria in the two halves as "isolated economies" (we'll justify the isolation from time 0 in a bit), and you'll see that, in the top half of the tree, Bob will (in equilibrium) move from his endowment of 6 units of the good in f_1^1 to only 5 units, while increasing his consumption in the two successor time-2 contingencies from 9 to $45/4$ apiece. Alice will go from 2 units to 3, decreasing her time-2 consumption to $27/4$ apiece. If we normalize the price of f_1^1 consumption to be 1, this takes prices of time-2 consumption of $2/9$ in each of the time-2 contingencies. And since we want π to be $1/2$ in $f_2 = \{\omega_1\}$, this means that we will set

$$r_{f_1^1} = 1, r_{\{\omega_1\}} = 4/9, \text{ and } r_{\{\omega_2\}} = 2/9.$$

In the bottom half of the tree, everything is symmetric but with the roles of Alice and Bob reversed, and we are going to have $\pi = (1, 1, 1)$, so we set

$$r_{f_1^2} = 1, r_{\{\omega_3\}} = 2/9, \text{ and } r_{\{\omega_4\}} = 2/9.$$

Let me slow this down, a bit. Suppose in the top half of the tree, r is given as above, the dividends of the two securities are as given, and Bob begins with y units of wealth (in the numeraire) in addition to his endowments. (We'll want to set prices in f_0 so he chooses not to trade between times 0 and 1, but we'll need to know the marginal value to him and to Alice of wealth in the two time-1 contingencies, so we carry this y along.) Bob's decision variables are how much of the two securities to hold leaving time 1; call them α_1 and α_2 as before. He'll clearly eat anything he doesn't immediately invest in securities, and he'll eat his endowment plus the dividends he collects at time 2, so his problem is to maximize

$$\ln(6 + y - 8\alpha_1 - 8\alpha_2) + \frac{1}{2} \left[\ln \left(9 + \frac{2\alpha_1 + 12\alpha_2}{4/9} \right) + \ln \left(9 + \frac{7\alpha_1 + 2\alpha_2}{2/9} \right) \right].$$

And, simultaneously, Alice is maximizing

$$\ln(2 + y' - 8\beta_1 - 8\beta_2) + \frac{1}{2} \left[\ln \left(9 + \frac{2\beta_1 + 12\beta_2}{4/9} \right) + \ln \left(9 + \frac{7\beta_1 + 2\beta_2}{2/9} \right) \right],$$

where y' is any extra money she has at time 1, and her portfolio weights are β_1 and β_2 . Solving this numerically for $y = y' = 0$ gives $\alpha_1 = 0.05$ and $\alpha_2 = 0.075$, providing Bob with 5 units of consumption in f_1^1 and 11.25 in each of $\{\omega_1\}$ and $\{\omega_2\}$, while Alice optimally takes $\beta_1 = -0.05$ and $\beta_2 = -0.075$, giving her 3 units of consumption in

f_1^1 and 6.75 in each of the two time-2 contingencies in the upper half of the tree. It is worth noting that, for Bob, the marginal utility he would get from more money in f_1^1 is $1/5$, while for Alice it is $1/3$. (Question: These are, respectively, Bob's and Alice's marginal utilities in their respective f_1^1 consumption levels; utility is logarithmic and Bob is at 5 units, while Alice is at 3. Perhaps in retrospect, it should be "obvious" to you that this is so. Why? If you don't know, think back to Chapter 3!)

A similar exercise in the bottom half of the tree (the contingency f_1^2 and its two successors $\{\omega_3\}$ and $\{\omega_4\}$ will show you that with the prices of q as given and for $r_{f_1^2} = 1$, $r_{\{\omega_3\}} = r_{\{\omega_4\}} = 2/9$, Bob (starting from his endowment of 2 in f_1^2 will take the position of 0 units of the first security and -0.125 units of the second, which gives him 3 units of consumption in f_1^2 and 6.75 in both time-2 contingencies, while Alice, starting from her endowment of 6, buys 0.125 units of the second security and none of the first, which leaves her at (5, 11.25, 11.25). And their respective marginal utilities of money are $1/3$ for him and $1/5$ for her.

To complete the construction of this EPPPE, we need to set a price for the good in period 1 so that neither Alice nor Bob want to buy or sell any of the securities. They are in symmetric positions: Each 1 unit of numeraire spend for purchasing either security (the securities have identical immediate returns) provides 0.5 units of numeraire next time. Given that prices next time of the consumption good are 1 apiece, and there is a $1/2$ probability of either contingency, 0.5 units of numeraire next period means a marginal gain in expected utility of $0.25(1/5 + 1/3) = 2/15$. So if the price of the consumption good in f_0 is $15/2$, neither Alice nor Bob want to take a nonzero position in either security, and markets clear.

Hence, we have the following *incomplete-markets EPPPE*:

$$r_{f_0} = 15/2, r_{f_1^1} = r_{f_1^2} = 1, r_{f_2} = \begin{cases} 4/9, & \text{if } f_2 = \{\omega_1\}, \text{ and} \\ 2/9, & \text{otherwise.} \end{cases}$$

$$q_{f_0} = (16, 16) \quad \text{and} \quad q_{f_1^1} = q_{f_1^2} = (8, 8),$$

where we are giving the prices of the two securities in order,

$$x^{\text{Bob}} = (1, 5, 3, 11.25, 11.25, 6.75, 6.75) \quad \text{and} \quad x^{\text{Alice}} = (1, 3, 5, 6.75, 6.75, 11.25, 11.25),$$

and trading plans as described above; in particular, no positions taken at date 0, but Alice and Bob using the two securities at date 1 to move their endowments around to get to the consumption plans just given.

As a final note on all this, in the Walrasian equilibrium and in the EPPPE of parts b and c of this problem, Alice and Bob both have an expected utility of 3.58351894, while in this EPPE, the each have expected utility of 3.51898042. This example is much more

“knife-edge” than Hart’s example (because there are enough securities around so that, barring the unhappy coincidence caused by our bad choice of π , markets will be complete), but it is another example of EPPPE for a given economy that are Pareto ordered.

■ 16.10. I’ll sketch the argument: The problem has already enumerated the states so that $d(J)$ is the largest of all dividends. Reordering as needed, we suppose they are enumerated so that $d(J) > d(J-1) > d(J-2) > \dots > d(1)$. Then, derivative security $J-1$ pays the positive dividend $d(J) - d(J-1)$ in contingency $\{\omega_J\}$ and zero in all other contingencies, while derivative security $J-2$ pays $d(J) - d(J-2)$ in $\{\omega_J\}$, $d(J-1) - d(J-2)$ in $\{\omega_{J-1}\}$, and zero in all the others. So a portfolio of 1 unit of s^{J-2} and $-(d(J) - d(J-2))/(d(J) - d(J-1))$ units of s^{J-1} pays a positive dividend of $d(J-1) - d(J-2)$ in contingency $\{\omega_{J-1}\}$,

$$[d(J) - d(J-2)] - \frac{(d(J) - d(J-2))}{(d(J) - d(J-1))}(d(J) - d(J-1)) = 0$$

in contingency $\{\omega_J\}$, and zero in all other contingencies. By similar (but increasingly complex) manipulations, we can form portfolios of the primitive security and the derivatives that give a strictly positive dividend in each individual time T contingency and zero in all the others; the key is that the matrix of dividends forms a triangular array. But then these portfolios, bought at time 0 and sold in particular f_t , constitute a way to transfer wealth from f_0 to each f_t (see the first half of the solution of Problem 16.6), and Lemma 16.10 applies.

■ 16.12. If the set of securities is $\mathcal{S}^{\text{FFCC}}$, then the only time the security connected to contingency f_t trades is at time 0. Hence, with $N-1$ securities, q is $N-1$ dimensional. And, if we specify π , its price according to equation (16.4) is π_{f_t}/π_{f_0} . So to the extent that π can be anything, so can q . Or put it this way: Take an arbitrary $q \in R_{++}^{N-1}$. Define $\pi \in R_{++}^N$ by $\pi_{f_0} = 1$ and $\pi_{f_t} = q_{s^{f_t}f_0}$, where s^{f_t} is the security in $\mathcal{S}^{\text{FFCC}}$ that pays in contingency f_t . For a Walrasian equilibrium (p, x) for the all-markets-at-once economy, let $r_{if_t} = p_{if_t}/\pi_{f_t}$, and the machinery of the chapter shows that, for the right choice of y , (r, q, x, y) will be an EPPPE.

■ 16.14. Suppose there are two periods, $t = 0, 1$, and one state of nature. Then there are two contingencies, f_0 and a single f_1 . If there are no securities, then the two contingencies are economically isolated from one another. Now suppose that, in each contingency, Alice and Bob face the situation that, in Hart’s example, they face in the two time-1 contingencies. There are then four EPPPE, and just as in Hart’s example, two are Pareto ordered.

■ 16.16. Since there is no obvious way to define “profit,” we’ll begin our analysis of this economy by answering the question: Suppose the firm decides on the production plan in which it purchases z units of input in contingency f_0 , generating $2z$ units of output in each of the two time-1 contingencies. What prices will equilibrate this market?

In precisely the manner of the analysis in the text, we set the price of the one consumption good to 1 in each contingency, so the only two prices are of the savings security, the price of which I'll call q , and the equity of the firm, whose price I'll call q' . We must specify how the firm finances its purchases of input and what it does with its output; here are my assumptions: The firm borrows money by selling the savings instrument; to get z units of input, it must sell z/q units of the savings instrument. It then pays back its loans at time 1, and pays out any money left over as a dividend. So holding a share of equity generates the cash flow of 0 at f_0 and $z(2 - 1/q)$ in either time-1 contingency. We hope that $2 \geq 1/q$ as, if not, the dividend is negative. If this happens, though, it will be clear that the best choice of z is $z = 0$, since raising z only lowers the value of the firm's equity, something neither Alice nor Bob want.

But, in this case, buying a unit of the firm's equity generates the cash flow

$$(-q', z(2 - 1/q), z(2 - 1/q))$$

in the three contingencies f_0, f_1^1, f_1^2 . Buying a unit of the savings instrument generates $(-q, 1, 1)$. So, to prevent the possibility of arbitrage, since the time-1 returns from the two securities are collinear, we must have in equilibrium that

$$\frac{z(2 - 1/q)}{q'} = \frac{1}{q}, \quad \text{or} \quad q' = z(2q - 1).$$

I could try to find the equilibrium prices by looking at Alice's and Bob's demands, to see where supply equals demand. But that's difficult, so I'll try a different tack. As long as the firm issues the same dividend in each time-1 contingency (which, so far, we are assuming), Alice and Bob are in symmetric situations. So their demands at time 0 must be the same, and they must end up with the same levels of consumption at time 0. Specifically, if the firm takes z units of the good to use as input, both Alice and Bob (in equilibrium) will be consuming $10.5 - z/2$ in contingency f_0 .

Moreover, since they have identical positions in the two security markets, they receive identical dividends and identical payouts from the savings security. They don't consume the same amount, because they have different endowments. But if the firm uses z units of input, it produces $2z$ units of output. Some of that goes to paying off the firm's debts, but those debts are owed (in equal measure) to Alice and Bob, who loaned the money to the firm; some is paid in the form of dividends. The point is, Alice and Bob each get z from the securities markets in both contingencies. So Alice's consumption in f_1^1 is $18 + z$, while in f_1^2 she consumes z ; Bob's is symmetric.

Hence we can compute Alice's marginal utilities at her equilibrium consumption levels in each of the three contingencies. And this drives the price of the savings instrument:

$$q = \frac{\text{MU}(f_1^1) + \text{MU}(f_1^2)}{\text{MU}(f_0)},$$

where by $MU(f_1^1)$ I mean $0.5/(1 + x_{f_1^1})$, $MU(f_1^2) = 0.5/(1 + x_{f_1^2})$, and $MU(f_0) = 1/(1 + x_{f_0})$; the consumption levels being those of Alice. Note why this must be true: At the equilibrium values, if Alice thinks of purchasing δ more of the savings instrument, the first-order impact on her overall utility will be

$$\delta[-qMU(f_0) + MU(f_1^1) + MU(f_1^2)],$$

so if the term inside the square brackets isn't zero, she won't be content wherever she is. In equilibrium, this gives q . (Of course, the same is true of Bob's marginal utilities, but since his situation is symmetric, he gives the same value of q .)

So, for instance, suppose the firm chooses $z = 0.7$. In equilibrium, Alice consumes $10.5 - 0.35 = 10.15$ at contingency f_0 , $18 + 0.7 = 18.7$ in f_1^1 , and $0 + 0.7 = 0.7$ in f_1^2 . Her marginal utilities are, respectively, 0.08968610, 0.02538071, and 0.29411765, so q must be 3.56240669.

And once we know q , we know how many units of the savings security the firm must sell (in this case, to get 0.7 units of the good, it must sell $0.7/3.56240669$ units; in general, the number is z/q), hence we know the size of dividend the firm will pay (in general, $2z - z/q$; in this case, 1.20350363), and so we know by arbitrage what is the value of q' , $z(2q - 1)$ in general, and 4.28736936 in this case. Alice's net positions (after time-0 trading) in the two securities are also determined: Because of her symmetry with Bob, she must buy half of the firm's debt, or $z/2q$ units of the savings security (in this case, 0.09824819 units), and she must hold half the firm's equity. Please note that at time 1, in either contingency, the firm pays back to Alice its loan of $z/2q$ and pays her a dividend of $0.5(2z - z/q)$, so her net proceeds from the securities markets are z , as they should be.

That is everything we need to know, for the cases where the firm purchases z units of input and liquidates all its output (to settle its debts and pay a dividend with the rest). In Table M16.1, I provide the equilibrium values (prices and utility levels for Alice and Bob) for $z = 0, 0.3, 0.7$, and 1. You see there that the firm's value of equity (the "price of a share") is largest when $z = 1$; not surprisingly, perhaps, if the firm is going to follow this sort of policy, its equity has largest value when it uses its valuable production facilities to the largest extent possible. And $z = 1$ also maximizes Alice's utility; since Bob is in a symmetric position, if these were the only production policies available to the firm, this is what Alice and Bob would want the firm to do.

Now suppose the firm contemplates producing some amount and, in one of the two contingencies, disposing of some of it. To match what is in the text, I'll assume that the firm will dispose of 0.01 units in contingency f_1^1 . I'll also assume that the firm purchases 1 unit of input, so its gross production (before disposal) is 2 units in both contingencies.

To find the equilibrium prices that go with this, I can't use the methods used previously in this problem, because Alice and Bob will no longer be in a symmetric posi-

	Value of z , the amount of input used by the firm			
	0	0.3	0.7	1
price of savings instrument	6.05263158	4.65942607	3.56240669	3.025
price of 1 share of the firm	0	2.49565564	4.28936936	5.05
Alice's and Bob's utility levels	3.91456652	4.04045242	4.16706294	4.242335

Table M16.1. Equilibrium Values if the Firm Produces and "Pays Out" the Same Amount in the Two Time-1 Contingencies. This table shows the equilibrium prices of the savings security and equity of the firm, if the firm chooses to input z and pay out, as a mixture of discharged debts and dividends, all of its output, in both time-1 contingencies, for $z = 0, 0.3, 0.7$, and 1. Since the situations of Alice and Bob are symmetric, it is easy to compute their consumption levels and security holdings; see the text.

tion. So I use a different logic. (This, by the way, is how I found the equilibrium values for the example in the text.)

Note first that, in this case, because the dividends will be different in the two time-1 contingencies, equity in the firm and the savings security provide enough "independent" security instruments to get complete markets.

Therefore, we know that the consumption allocation portion of the equilibrium must give a Pareto-efficient division of the amounts available for consumption: 20 in f_0 , 19.99 in f_1^1 , and 20 in f_1^2 . We know (from back in Chapter 8) how to find Pareto-efficient divisions of a fixed pile of goods—you pick weights for Alice and Bob and maximize their weighted utilities. If we give Alice a weight of λ and Bob a weight of 1, this means that ratio of Alice's marginal utility to Bob's in each separate contingency must be equalized (the three ratios must be the same), and if we pick a value of λ , it is a matter of simple algebra (given these simple utility functions) to find the corresponding efficient allocation.

So pick a value for λ and find the allocation that goes with it. *If* this is the equilibrium allocation, it determines both the price of the savings security and then the price of the firm's equity: For the price of the savings security, use the formula given above. Then use the price of the savings security to compute how much the firm must borrow and, hence, what dividends the firm will pay in the two contingencies. And then use the

formula

$$\text{Price of the firm's equity in } f_0 = \frac{\text{MU}(f_1^1)d_{\sigma f_1^1} + \text{MU}(f_1^2)d_{\sigma f_1^2}}{\text{MU}(f_0)}.$$

Here, $d_{\sigma f_1^i}$ is the dividend paid by the firm in contingency f_1^i and the MUs are Alice's marginal utilities at the consumption allocation we're trying. (Because of the efficiency condition, we'd get the same value if we used Bob's marginal utilities.)

Once we know q , the price of the savings security, and q' the price of the firm's equity, we can back out what Alice must hold in her portfolio, to get to the consumption levels in the two time-1 contingencies we are trying. And we can compute, based on that portfolio, what is her consumption in f_0 . If her implied consumption in f_0 matches what we first computed for her consumption, then (you can show) we have equilibrium values; the trick then is to adjust the weight on Alice so that everything balances.

Based on this, the (approximate) equilibrium I found is shown in Table M16.2. Note that the utility levels for both Alice and Bob are significantly higher than in the best case if the firm doesn't engage in a bit of destruction. And, of course, Alice will, *ex post*, tell the firm not to bother destroying any of the good in contingency f_1^1 , which will bankrupt Bob.

Price of savings instrument	1.000227376
Price of equity of firm	0.995452478
Units of the savings security borrowed by the firm	0.999772676
Dividend paid by the firm in second time-1 contingency	1.000227324

	Alice	Bob
Positions in savings instrument	-1790.90833	1791.908099
Positions in equity of firm	1800.500046	-1799.500046
Consumption time 0	10.00101741	9.998982594
Consumption time 1 state 1	9.996016943	9.993983057
Consumption time 1 state 2	10.00101741	9.998982594
Utility of outcome	4.79574925	4.79537718

Table M16.2. Equilibrium Values if the Firm Uses One Unit of Input and Then Destroys a Bit of Output in Contingency f_1^1 . This table shows the equilibrium values if the firm chooses to input 1 and then pay out, as a mixture of discharged debts and dividends, all of its output in contingency f_1^2 but 0.01 less than all its output in contingency f_1^1 . Note that the dividend paid in contingency f_1^1 is 0.01 less than that paid in f_1^2 . The values here are numerically derived and are approximate.